

Tracking a Plane Wave Direction using Sparse Arrays with an Extended Kalman Filter

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Abstract—This work focuses on estimating the direction of arrival of a plane wave from a moving source using time domain snapshots captured by a linear array. We employ an extended Kalman filter by linearizing the time domain measurement model. We consider the performance of a full array, which serves as a benchmark for comparison with three popular sparse arrays: coprime, nested, and shift-invariant sparse arrays. The motivation for using such sparse arrays is to reduce overall cost, sensor count, and computation burden. The simulation results demonstrate that sparse arrays achieve comparable accuracy to full arrays of equivalent aperture. Among the sparse arrays studied, the coprime array exhibits the highest performance, with a mean squared error closest to that of the full array.

Index Terms—Coprime array, direction of arrival (DOA), extended Kalman filter(EKF), nested array, shift-invariant sparse array (SISA), sparse arrays

I. INTRODUCTION

A uniform linear array (ULA) is the most commonly studied array structure for estimating the direction of arrival (DOA) of plane waves [1]–[4]. In a ULA, sensors are positioned along a straight line with an intersensor distance of $\lambda/2$, where λ is the wavelength of the impinging waves [1]. When intersensor spacing exceeds $\lambda/2$, aliasing artifacts may occur. However, careful design of sparse arrays can mitigate these artifacts. Arrays with an average intersensor spacing greater than $\lambda/2$ are classified as sparse arrays. For this study, we focus exclusively on sparse arrays arranged on a uniform $\lambda/2$ grid along a line. These sparse linear arrays (SLAs) allow for a reduction in the number of sensors by increasing the spacing. Consequently, SLAs offer lower costs and require less data computation compared to ULAs [5], [6]. Additionally, the sparse positioning of sensors in SLAs results in lower mutual coupling effects [7], [8]. These advantages make sparse arrays desirable for various applications [9]–[12], and they come in different types [2], [13], [14].

Coprime and nested arrays are constructed by interleaving two undersampled ULAs (ULAs with an intersensor spacing greater than $\lambda/2$) [7], [15]. The design parameters for these arrays include sampling factors, number of sensors, and array lengths. By adjusting these parameters, various types of sparse linear arrays (SLAs) can be created. A recent addition to the category of sparse arrays is the shift-invariant sparse array (SISA), which encompasses both coprime and nested

arrays [16]–[18]. In this study, we analyze and compare the performance of three types of sparse arrays: (1) coprime, (2) nested, and (3) SISA.

For signal estimation and tracking, we apply the extended Kalman filter (EKF). The Kalman filter, introduced by R.E. Kalman in 1960, stands as one of the most prominent and widely used signal parameter estimators [19]. The Kalman filter is a sequential minimum mean squared error (MMSE) estimator and it allows the unknown parameters to evolve in time via a dynamical model [20], [21]. A Kalman filter can be regarded as a generalization of the Wiener filter since a Kalman filter does not require the signals to be stationary and scalar. Kalman filter provides the estimates of the parameters of interest by incorporating data from past and present [20]. When the signal and noise follow a jointly Gaussian distribution, the Kalman filter represents the optimal MMSE estimator. Even in scenarios where this assumption is not met, it remains the optimal linear MMSE (LMMSE) estimator. However, the Kalman filter is inherently limited to linear signal and measurement models, a constraint not met by many practical systems. In such cases, an extended version (i.e., EKF), is employed. The EKF leverages a first-order Taylor series approximation to linearize non-linear signal and measurement models, thus extending the applicability of the traditional Kalman filter concept [21], [22]. In this study, we utilize time domain snapshots received by a linear array as measurements. Since these measurements are non-linear in the parameter of interest, we apply an EKF for direction of arrival (DOA) estimation.

This paper conducts a comparative analysis of SLAs employing an EKF for tracking the DOA of a plane wave. The study centers on three distinct SLAs (coprime, nested, and SISA), each with an identical number of sensors (6) and aperture. A full ULA comprising 10 sensors is the benchmark for comparison.

The rest of this paper is structured as follows. Section II describes the model of the signal and the estimator (EKF). Section III explains and illustrates the array models. Section IV presents and discusses the simulation results. Section V sums up the study.

II. SIGNAL MODEL AND ESTIMATOR

We consider the received signal as a sinusoid with a known amplitude α and an unknown direction θ . We further assume that the signal direction changes over time, and therefore, we employ the EKF to track the evolving direction cosine $\cos(\theta)$. In the following sections, the boldface lowercase symbols represent vectors, while boldface uppercase symbols denote matrices. Consistent with the notation style in reference [21], we denote $s[n]$ as $\cos(\theta[n])$. Thus, the received signal at the i^{th} sensor at time n is given by

$$\mathbf{x}[n] = \alpha \cos(\omega_0 n + \pi s[n]i) + w_i[n], \quad (1)$$

where ω_0 is the temporal radian frequency and $w_i[n]$ is the white Gaussian noise (WGN) at sensor i and time n . We can also express (1) as

$$\mathbf{x}[n] = \alpha \begin{bmatrix} \cos(\omega_0 n + \pi s[n]0) \\ \cos(\omega_0 n + \pi s[n]1) \\ \vdots \\ \cos(\omega_0 n + \pi s[n](L-1)) \end{bmatrix} + \mathbf{w}[n], \quad (2)$$

where L is the number of sensors in the array. Since (2) is non-linear in the unknown $s[n]$, an EKF needs to be applied instead of the traditional Kalman filter. Comparing (2) with (13.66) from [21], we can rewrite (2) as

$$\mathbf{x}[n] = \mathbf{h}(s[n]) + \mathbf{w}[n]. \quad (3)$$

In this study, we assume that the unknown direction cosine $s[n]$ is a scalar that can be modeled as an auto-regressive process of order 1:

$$s[n] = A s[n-1] + u[n], \quad (4)$$

where $u[n]$ is WGN with variance Q . The state equation in (4) is linear and hence Taylor series approximation is not required for this equation.

To linearize $\mathbf{h}(s[n])$ in (3), we differentiate with respect to $s[n]$ to obtain

$$\frac{\partial \mathbf{h}(s[n])}{\partial s[n]} = -\alpha \begin{bmatrix} \pi 0 \\ \pi 1 \\ \vdots \\ \pi(L-1) \end{bmatrix} \odot \begin{bmatrix} \sin(\omega_0 n + \pi s[n]0) \\ \sin(\omega_0 n + \pi s[n]1) \\ \vdots \\ \sin(\omega_0 n + \pi s[n](L-1)) \end{bmatrix} \quad (5)$$

where $\odot$ represents the Hadamard or element-wise product. Denoting the sensor indices as $\mathbf{i} = [0 \ 1 \ \dots \ L-1]^T$, we can rewrite (2) as $\mathbf{x}[n] = \alpha \cos(\omega_0 n + \pi s[n]\mathbf{i}) + \mathbf{w}[n]$ and (5) as

$$\frac{\partial \mathbf{h}(s[n])}{\partial s[n]} = -\alpha \pi \mathbf{i} \odot \sin(\omega_0 n + \pi s[n]\mathbf{i}). \quad (6)$$

Setting $\mathbf{i} = [0 \ 1 \ \dots \ 9]^T$ yields a full ULA with 10 sensors. The SLAs can also be specified in a similar manner by changing the sensor indices. For instance, the coprime array, nested array, and SISA in Fig. 1 correspond to

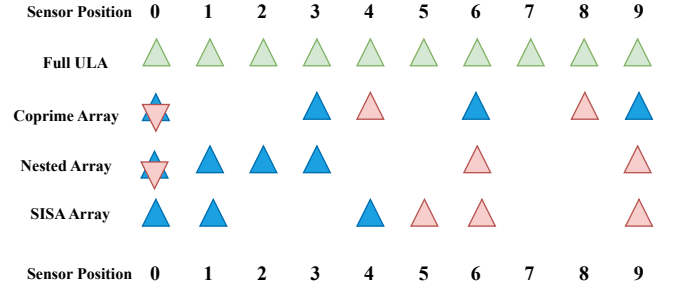


Fig. 1. Sensor positions for the ULA, coprime, nested, and SISA arrays (blue and red colors indicate subarray 1 and subarray 2, respectively)

$\mathbf{i} = [0 \ 3 \ 4 \ 6 \ 8 \ 9]^T$, $\mathbf{i} = [0 \ 1 \ 2 \ 3 \ 6 \ 9]^T$, and $\mathbf{i} = [0 \ 1 \ 4 \ 5 \ 6 \ 9]^T$, respectively.

The linearized observation equation is [21]

$$\mathbf{x}[n] = \mathbf{H}[n]s[n] + \mathbf{w}[n] + (\mathbf{h}(\hat{s}[n|n-1]) - \mathbf{H}[n]\hat{s}[n|n-1]), \quad (7)$$

where $\mathbf{H}[n] = \left. \frac{\partial \mathbf{h}(s[n])}{\partial s[n]} \right|_{s[n]=\hat{s}[n|n-1]}$ and $\hat{s}[n|n-1]$ is the predicted value of $s[n]$ based on $\{x[0], x[1], \dots, x[N-1]\}$. The EKF equations are [21]

- Prediction:

$$\hat{s}[n|n-1] = A\hat{s}[n-1|n-1]; \quad (8)$$

- Minimum Prediction MSE:

$$M[n|n-1] = A^2 M[n-1|n-1] + Q; \quad (9)$$

- Kalman Gain Matrix:

$$\mathbf{K}[n] = M[n|n-1]\mathbf{H}^T[n] \times (\mathbf{C}[n] + M[n|n-1]\mathbf{H}[n]\mathbf{H}^T[n])^{-1}; \quad (10)$$

- Correction:

$$\hat{s}[n|n] = \hat{s}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{h}(\hat{s}[n|n-1])); \quad (11)$$

- Minimum MSE:

$$\mathbf{M}[n|n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1], \quad (12)$$

where $\mathbf{C}[n]$ is the covariance matrix of $\mathbf{w}[n]$, $\hat{s}[n-1|n-1]$ and $M[n|n-1]$ are the estimated value of $s[n-1]$ and the MSE based on $\{x[0], x[1], \dots, x[N-1]\}$ while $M[n|n]$ is the MSE based on $\{x[0], x[1], \dots, x[N]\}$.

III. SPARSE ARRAYS

The full ULA considered in this work consists of 10 sensors positioned from 0 to 9, as illustrated in Fig. 1. Each sparse array considered in this study comprises 6 sensors and spans the same aperture as a full ULA with 10 sensors. We provide a brief description of the sparse arrays next.

A. Coprime Array

A linear coprime array is formed by interleaving two linear subarrays whose undersampling factors are coprime integers

[6], [15], [23]–[25]. We denote the two subarrays as $\mathbf{S}_{1c}$ and $\mathbf{S}_{2c}$, respectively. The subarray $\mathbf{S}_{1c}$ has a total of M_e sensors and an undersampling factor of N , while $\mathbf{S}_{2c}$ has N_e sensors and an undersampling factor of M ; and M and N are coprime integers. We set $N = 3$ and $M = 4$, $M_e = 4$, and $N_e = 3$. The sensor positions of the coprime array are shown in Fig. 1.

B. Nested Array

A linear nested array is also formed by interleaving two subarrays. In a nested array, the first subarray $\mathbf{S}_{1n}$ has a sampling factor of 1 and comprises M sensors, while the second subarray $\mathbf{S}_{2n}$ consists of N sensors with an undersampling factor of M [6], [26]. The sensor positions of a nested array are also illustrated in Fig. 1.

C. Shift-invariant Sparse Array (SISA)

A shift-invariant sparse array is an array where one subarray can be obtained by applying a spatial shift to another subarray. The two subarrays may partially overlap and the union of the two subarrays is the overall SISA. As illustrated in Fig. 1, the first subarray consists of the sensors at $\psi_1 = [0 \ 1 \ 4]$; and applying a spatial shift of $v = 5$ yields the second subarray with the sensors at $\psi_2 = [5 \ 6 \ 9]$.

IV. SIMULATION RESULTS

We applied the EKF to track the direction cosine of a plane wave using the data measured by the full ULA and the sparse arrays described in Section III. In all experiments, we used $\omega_0 = 0.6\pi$, $N = 200$, $A = 0.95$, $Q = 0.01$, $\mathbf{C}[n] = 2\mathbf{I}$, $\hat{s}[-1|-1] = E\{s[n]\} = 0.5$, and $M[-1|-1] = 100$.

Fig. 2, Fig. 3, Fig. 4, and Fig. 5 illustrate the simulation results for the ULA, coprime array, nested array, and SISA, respectively. The top panels in Fig. 2 to Fig. 5 depict the true and estimated values of the direction cosine in one trial. The bottom panels in the figures plot the corresponding Bayesian MSE (B_{mse}). We computed the average B_{mse} over 500 trials for each array. Fig. 6 plots the average B_{mse} . All four arrays track the direction cosine reasonably well. As expected, the full ULA exhibits the highest accuracy as it has the lowest average B_{mse} . All sparse arrays exhibit degradation in performance compared to the full ULA. Among the three sparse arrays, the coprime array outperforms the other arrays. The nested array shows the highest average MSE.

V. CONCLUSION

We investigated the efficacy of an EKF for tracking the direction cosine of a plane wave, modeled by an autoregressive process of order 1. We applied the EKF to data collected from a full array comprising 10 sensors and comparing its performance against sparse arrays with an equivalent aperture but comprising only 6 sensors. Specifically, we explored coprime, nested, and SISA sparse arrays, discerning varying levels of performance across these configurations. Comparison of the average MSEs revealed that the full ULA outperforms

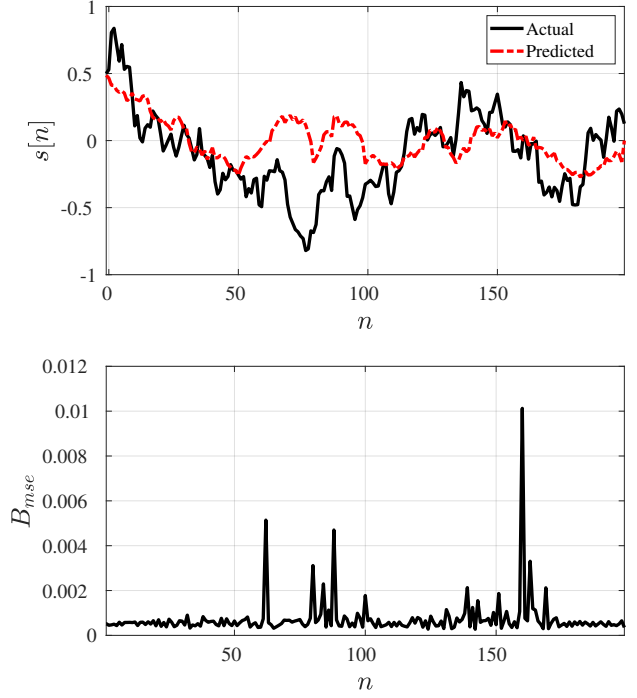


Fig. 2. Top: Actual and estimated values of the direction cosine for the full ULA. Bottom: MSE

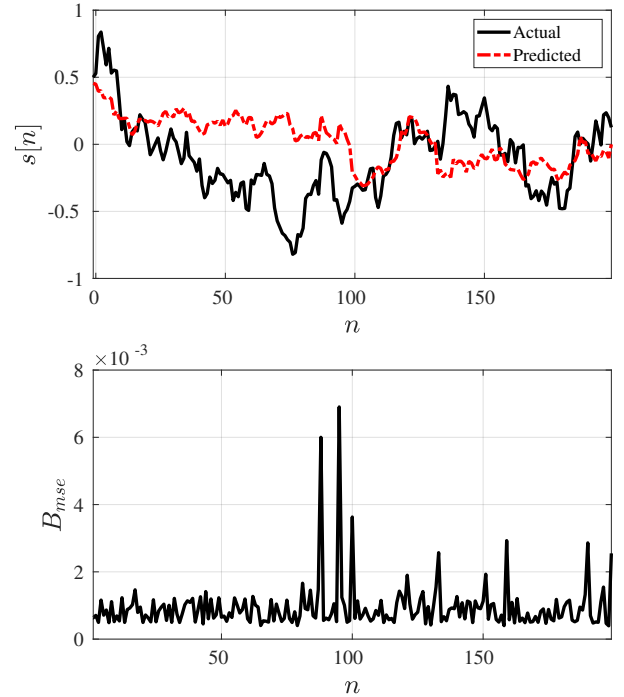


Fig. 3. Top: Actual and estimated values of the direction cosine for the coprime array. Bottom: MSE

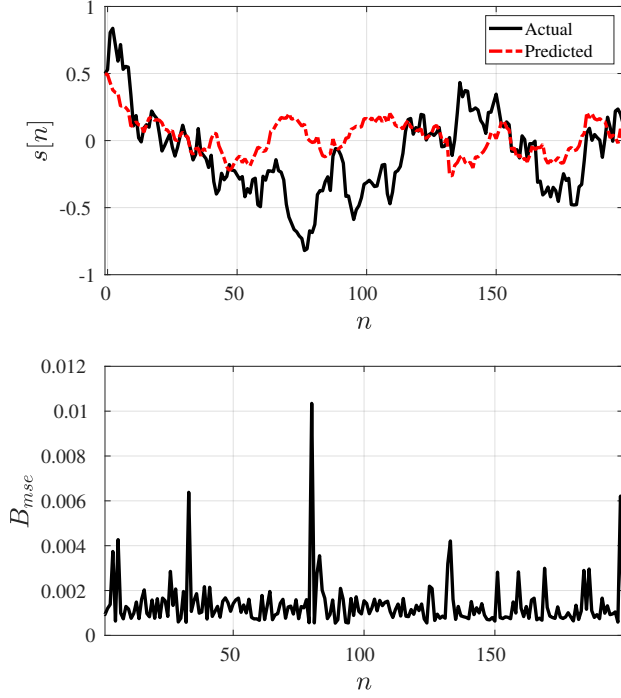


Fig. 4. Top: Actual and estimated values of the direction cosine for the nested array. Bottom: MSE

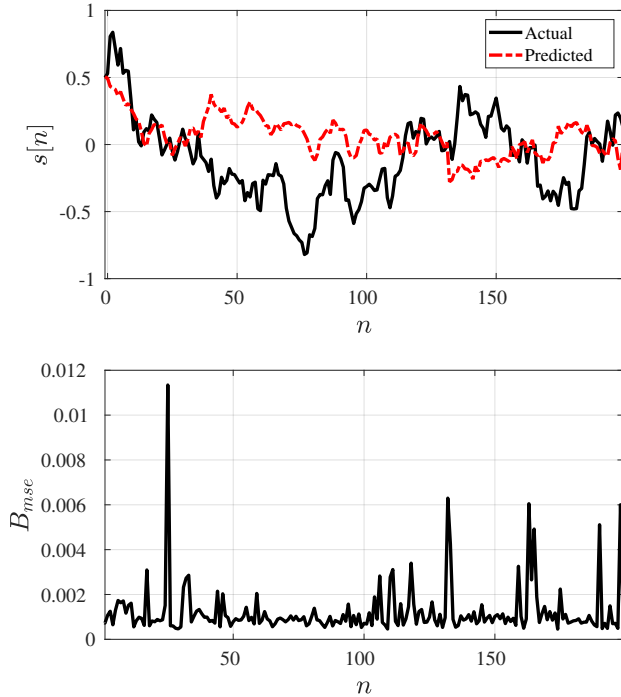


Fig. 5. Top: Actual and estimated values of the direction cosine for the SISA. Bottom: MSE

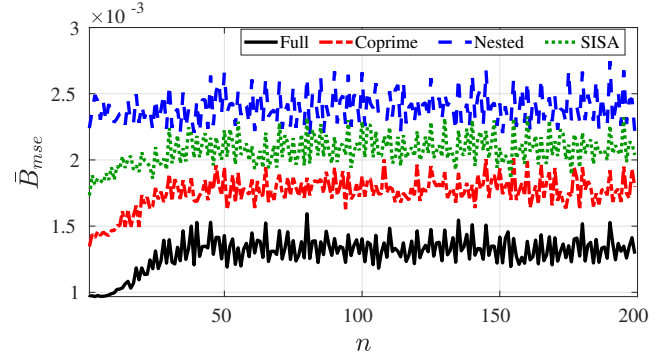


Fig. 6. Comparison of the average MSE of the four arrays.

the sparse arrays, as expected. Among the sparse arrays, the coprime array exhibited the lowest error while the nested array registered the highest error.

Future directions include analyzing further sparse array geometries and extending the study to tracking both angles of elevation and azimuth using two-dimensional arrays.

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